

Schedule to Section 63 of the Bharatiya Sakshya Adhiniyam, 2023

Part B — Technical Expert Certification (Section 79A, IT Act authorised expert)

Expert Name & Designation	Dr. Authorized Forensic Examiner — Digital Forensic Examiner (Sec 79A IT Act)
Expert ID / Licence	IT-79A-IND-2026-XXXX
Verification Workstation Serial	WS-IND-FORENSIC-9912
Verification MAC	00:1A:2B:3C:4D:5E
Independent SHA-256 Confirmation	a665a45920422f9d417e4867efdc4fb8a04a1f3fff1fa (verified match)
Date & Time of Examination (IST)	01-06-2026 19:42:19 IST

Forensic Analysis Summary

The electronic record identified above was extracted using write-protected forensic procedures. Structural integrity of the bitstream was preserved. No transcoding or lossy re-encoding was applied during acquisition.

- Spatial CNN manipulation probability: **0.9400**
- 3D face mesh landmark variance: **0.082300**

- Lip-sync alignment error: **127.50 ms**
- TTS synthetic voice probability: **0.8900**
- Spectrogram pitch mismatch ratio: **0.4100**
- Anti-spoofing NN confidence (bona fide): **0.1200**

Resolved Subject: Shri Example Politician (ID: candidate-042, fused similarity: 1.0000) — source: hybrid

Grounds for Deepfake Classification

Summary: Media classified as likely synthetically manipulated based on 12 forensic indicator(s), including 4 high-severity finding(s).

- **Audio-Video Sync (critical):** Audio–video lip movement is misaligned, consistent with synthetic dubbing or post-hoc voice replacement rather than naturally captured speech. (*lip_sync_alignment_error_ms: 127.5 ms*)
- **Facial Manipulation (high):** Spatial facial inconsistencies detected across frames suggest GAN or deepfake-based facial manipulation. (*spatial_cnn_manipulation_probability: 94.00%*)

- **3D Face Geometry (medium):** 3D face mesh landmarks are unstable across the temporal sequence, indicating inconsistent facial geometry typical of synthetic faces.
(*face_mesh_landmark_variance: 0.0823*)
- **Synthetic Voice (high):** The voice track exhibits characteristics consistent with text-to-speech or neural voice cloning rather than a live recording.
(*tts_synthetic_probability: 89.00%*)
- **Voice Authenticity (medium):** Pitch contour and spectrogram features do not match natural speech patterns for the apparent speaker, suggesting audio synthesis or splicing.
(*spectrogram_pitch_mismatch_ratio: 41.00%*)
- **Anti-Spoofing (high):** Anti-spoofing neural network assigns low confidence to bona fide speech, indicating the audio is likely non-genuine or heavily processed.
(*anti_spoofing_nn_confidence: 12.00%*)
- **Lighting Consistency (medium):** Lighting on the face and background shifts unnaturally between frames, suggesting compositing or generative synthesis rather than a single capture. (*illumination_inconsistency_score: 72.00%*)
- **Shadow Geometry (medium):** Shadow direction and softness are inconsistent with the scene light source, a common artifact in deepfake and face-swap videos.
(*shadow_geometry_inconsistency_score: 68.00%*)
- **Color Grading (medium):** Skin tone and color grading do not remain consistent across frames; color boundaries and shadows appear imperfect relative to natural video.
(*color_grading_anomaly_score: 61.00%*)

- **Temporal Stability (medium):** High-frequency flicker and temporal instability detected at facial boundaries, consistent with frame-wise generative manipulation.
(*temporal_flicker_score: 58.00%*)
- **Facial Boundaries (medium):** Visible artifacts at the jawline, hairline, or face perimeter suggest blending of a synthetic face onto an original body.
(*facial_boundary_artifact_score: 64.00%*)
- **Analyst Observation (medium):** Visible jawline blending; lighting on face inconsistent with background.

I certify that the above analysis was conducted on the hash-verified original and that the opinions expressed are based on validated multi-modal deepfake detection models.

Expert Signature: _____ Date: 01-06-2026